CS4437 · EXPLAINABLE AI · GROUP 8

Do explanations drift before accuracy does?

Watching a frozen electricity-price model through the 2026 oil shock

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The Strait of Hormuz
The only sea route out of the Persian Gulf.
About 33 km wide at its narrowest.
IRAN
Strait of Hormuz
Persian Gulf
Gulf of Oman
ARABIAN PENINSULA
Image: via USNI News, 27 Mar 2026

28 FEBRUARY 2026

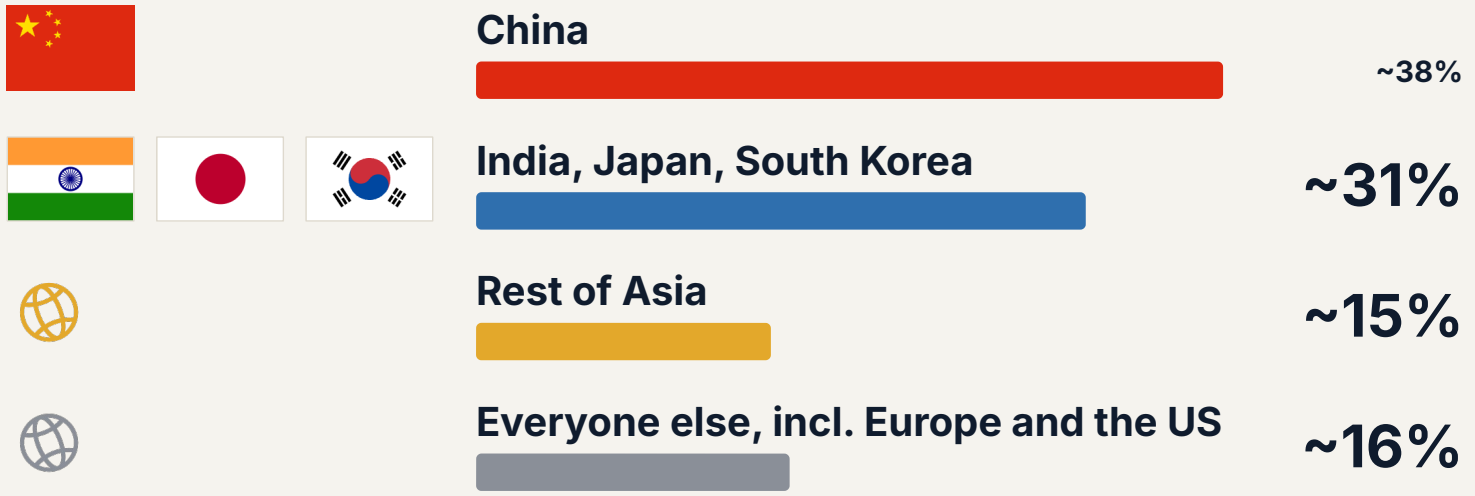
About a fifth of the world's oil goes through one strait. It closed.

~20%

of the world's oil use: about 20 million barrels a day

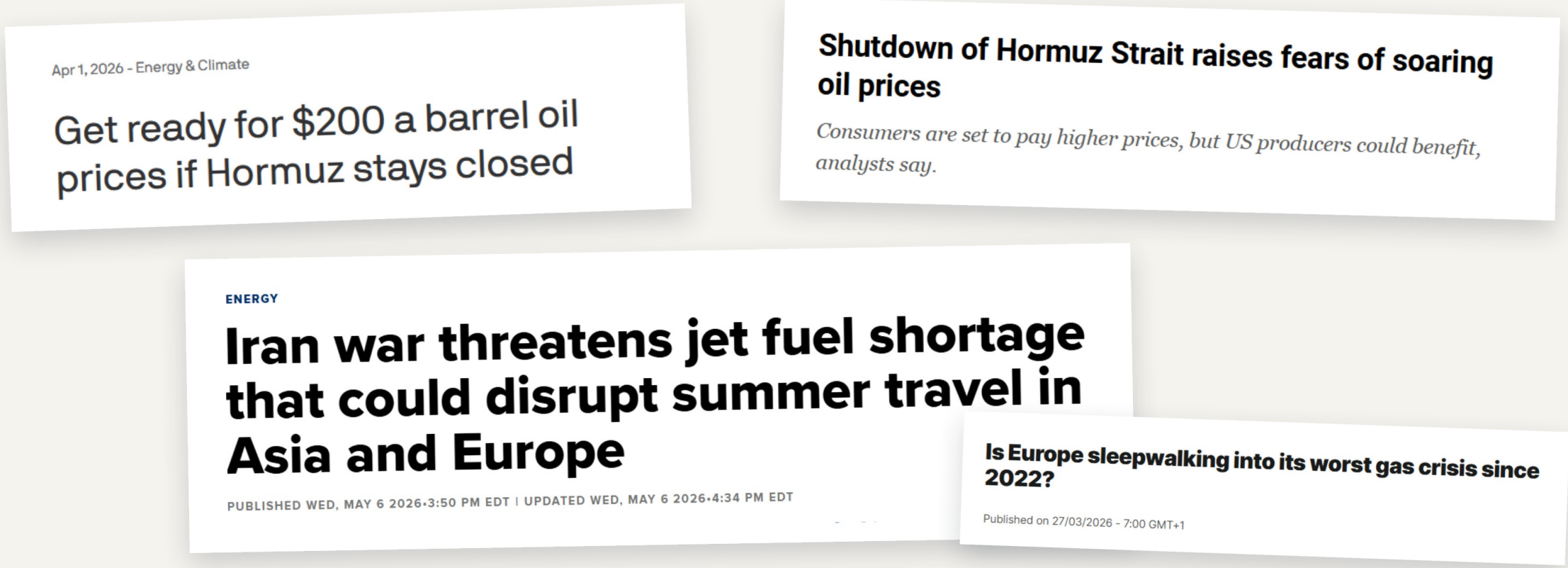
Europe buys little of it directly. But oil is priced globally, so **everyone pays more**.

WHERE THAT OIL GOES



US EIA, June 2025. Crude oil through Hormuz: Asia 84%, the four named countries 69% (2024); China's share from Q1 2025. Approximate.

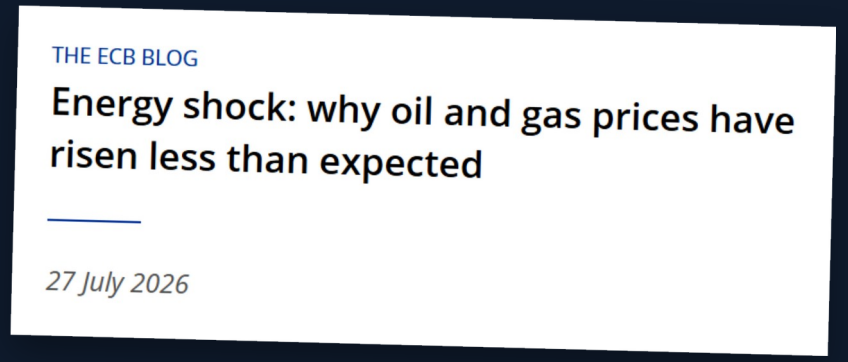
What we were told to expect



Al Jazeera, 3 Mar 2026 · Axios, 1 Apr 2026 · Euronews, 27 Mar 2026 · CNBC, 6 May 2026

WHAT ACTUALLY HAPPENED

The worst predictions didn't come true.


THE ECB BLOG

Energy shock: why oil and gas prices have risen less than expected

27 July 2026

So what did the forecasting models see?

Did their reasoning shift **before** their forecasts went wrong?

ECB Blog, 27 Jul 2026: oil peaked at just over +50%, where past shocks this size implied up to +105%

So we built our own model to find out

DATA



German electricity

Hourly prices, plus wind, solar and demand forecasts

PREDICT



LightGBM

Forecasts tomorrow's 24 hourly prices

EXPLAIN



SHAP

Which inputs each forecast relied on

MEASURE



Weighted tau

How much that changed, every two weeks

LightGBM + SHAP on German electricity prices follows Trebbien et al. 2023, so we start from a setup known to work.

We modelled electricity prices, not oil. Why?

Oil price: too obvious

The shock is the oil price itself. A model of it would obviously break, and that teaches us nothing.

Electricity price: one step removed

It feels the shock indirectly, through fuel costs. That's how most real systems get hit.

Plenty of open data

Germany publishes a price for every hour, free to use: 336 prices per fortnight.

The link is proven

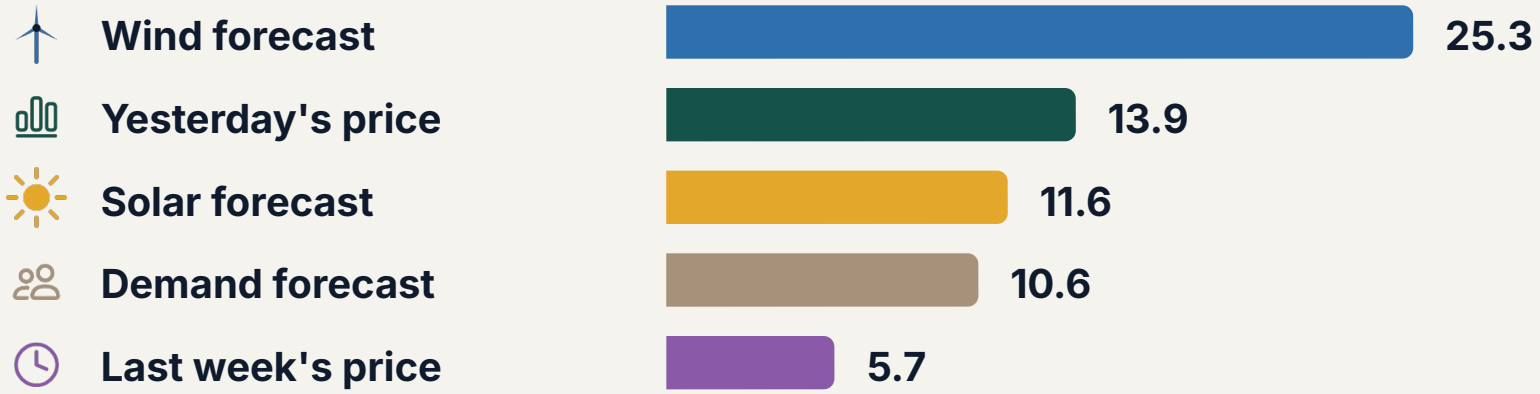
Fuel prices are among the top drivers of German power prices (Trebbien et al. 2023).

Oil and gas prices rise → Gas power plants cost more to run → The priciest plant needed sets everyone's price → **Electricity price rises**

What the model relies on

> ⚠ October is autumn. Keep that in mind for the results.

SHAP measures how many €/MWh each input moved each forecast up or down. Averaged over **October 2025**, our "before" snapshot that every later fortnight is compared against:



Everything else (temperature, gas power, hour, day, holiday): under 2 each.

One rule: only inputs known before the noon auction the day before, so the model never sees the answer.

Blind model, October 2025 · Data: SMARD, Open-Meteo (CC BY 4.0) · SHAP: Lundberg & Lee 2017; TreeSHAP: Lundberg et al. 2020

How we ran the experiment

1

Train

Jan 2024 – Sep 2025

Learns from 21 months of German prices.

2

Snapshot

Oct 2025

Check it's accurate. Record the "before" ranking.

Average gap from the real price:

14.5 €/MWh

vs **43.5** for just copying last week's price

3

Measure normal wobble

Nov 2025 – mid-Feb 2026

No shock yet. How much does the ranking move on its own? That sets the warning line.

4

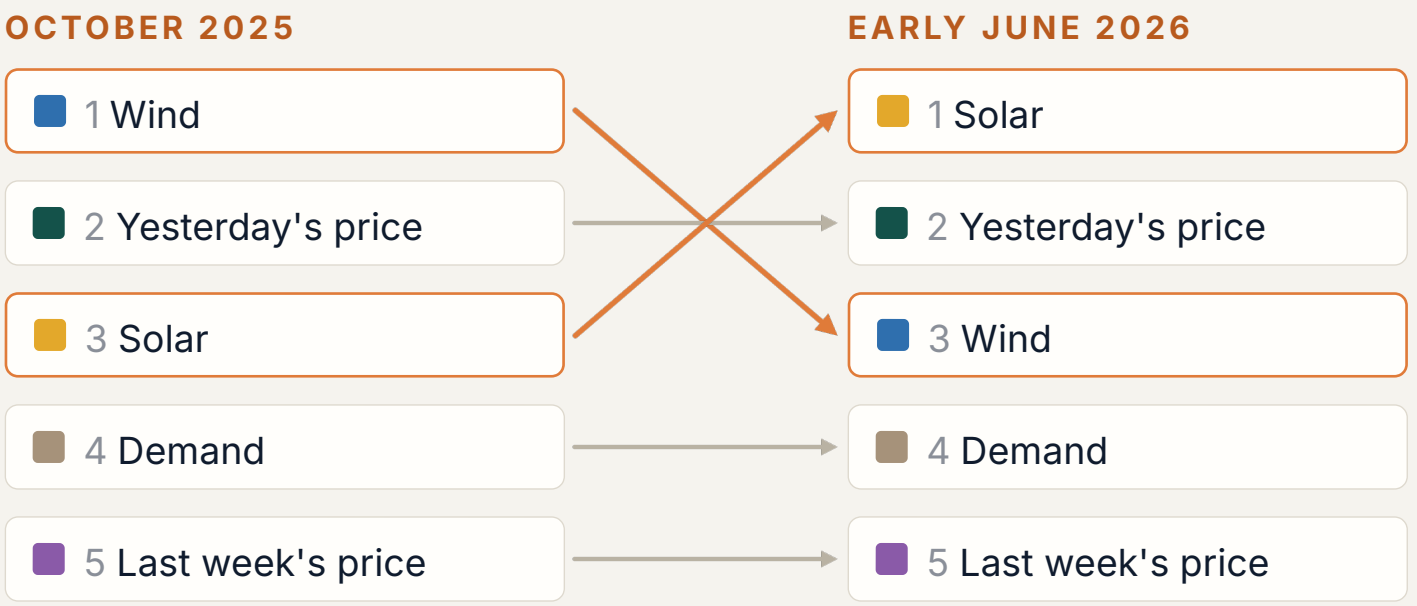
Watch

From the shock, 28 Feb 2026

Forecast every day. Every 2 weeks: compare the ranking with October, and check the error.

Trained once, then frozen. No retraining, so any change in its reasoning comes from the world changing, not from new training.

Measuring how far the ranking moves



MEASURE

Weighted Kendall's tau: 1 = identical ranking. Swaps near the top count more.

WARNING

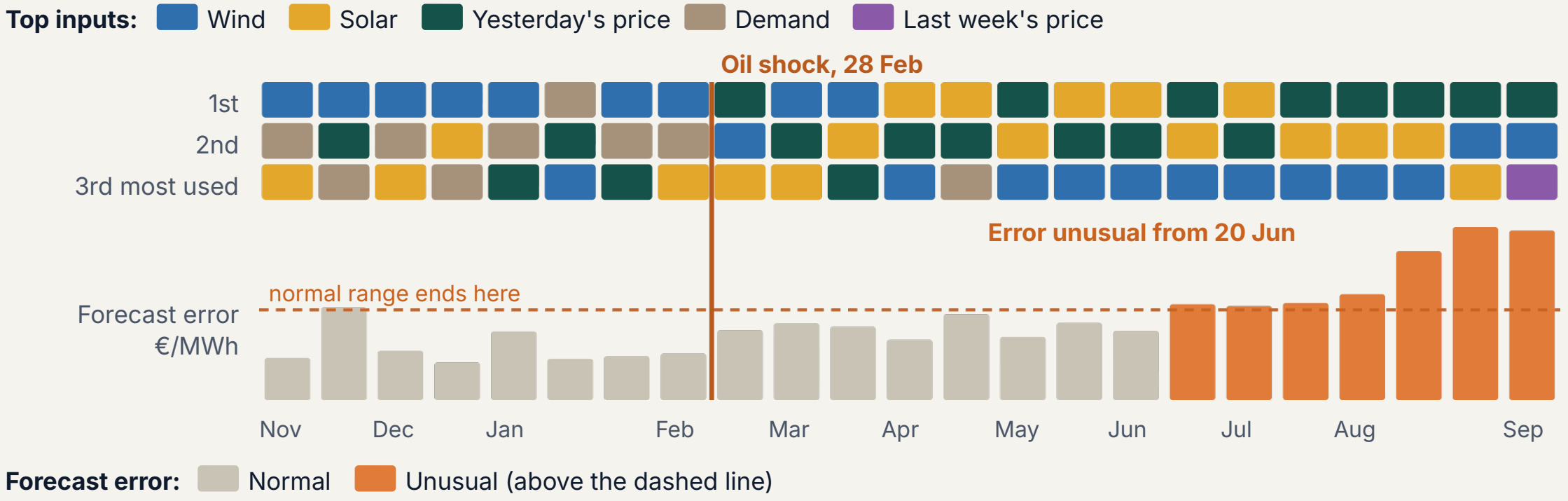
Line at 0.66: 2 SD below tau's pre-crisis normal. June scores 0.72, so no warning.

Wind and solar swap places at the top. That's summer, not the shock, and it pulls tau down to 0.72.

Example: the blind model's real fortnight from 6 June 2026. Baseline: 8 pre-crisis fortnights. Vigna 2015 (weighted tau).

The reasoning follows the calendar, not the shock

Blind model. Each column is one fortnight: its top three inputs, and its forecast error.



The ranking moved, but gave no warning

15/15

fortnights after the shock had yesterday's price in the top three

Before the shock: only 4 of 8.

Yesterday's price took over

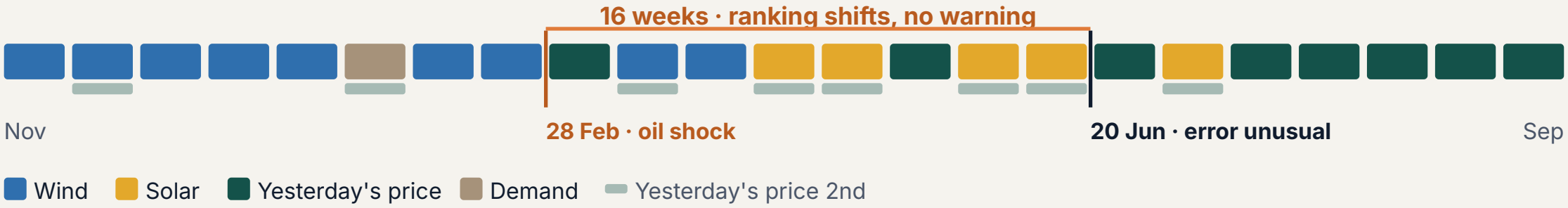
In the first fortnight after the shock it was the top input for the first time. From mid-July it led every fortnight.

Demand dropped out

In the top three in all 8 fortnights before the shock, but in just 1 of 15 after it.

But no early warning

Wind was back on top the next fortnight. Tau never crossed its warning line before the error did, on 20 June.



Each block is one fortnight's top input (mean |SHAP|), blind model. Pre-shock windows are all autumn and winter, so season and shock overlap.

What if the model could see oil?

1 · IT LOOKED LIKE A WIN

56 **days early**

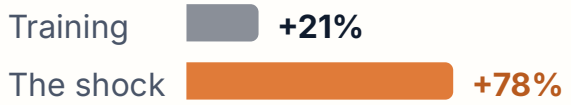
We added the oil price to the model. Its tau crossed the warning line on **25 April**, 56 days before the error did.

2 · BUT IT WASN'T OIL

Solar jumped to 1st

Our October snapshot is autumn, so **spring alone looks like drift**. Oil sat 11th of 13. Compare with spring 2025 instead and the warning disappears.

3 · WHY OIL STAYED QUIET



Its biggest monthly oil rise in training was 21%. Tree models can't go past what they've seen, so a 78% rise looked like 21%.

What it taught us: check a warning against the season before trusting it, and a model can't explain a shock it never saw.

Oil model = blind model + 30-day Brent change (FRED). Oil rank: mean |SHAP|, fortnight from 25 Apr 2026.

What our setup can't tell us

Rank, not size



Yesterday's price pushed 2.5× harder, but tau only sees it move up one place.

A shaky warning line



"Normal wobble" comes from just 8 calm fortnights, so the line itself is uncertain. Mid-January, with no crisis, already crossed it: a false alarm.

October did two jobs

When to stop training + **How accurate it is**

Using one month for both makes our 14.5 €/MWh error slightly optimistic.

One event, one market

1 shock 1 market 2 models

No early warning here doesn't rule one out for other shocks, markets or models.

TAKEAWAY

Explanation drift isn't a free early warning. Seasons can move it more than a shock.

For monitoring: watch explanations beside error, not instead of it.

Next: same-season baselines, a trend-based detector, more shocks.

Questions?

Lundberg & Lee 2017 · Lundberg et al. 2020 · Ke et al. 2017 · Trebbien et al. 2023 · Mougan et al. 2025 · Vigna 2015 · Code and data: github.com/matthewb62/cs4437-xai-group-project